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MANIFETCH

A Unified Web-Based Platform for Real-Time NICU Signal Monitoring

FINAL REPORT

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https://github.com/manifetch-lab/manifetch.git

1. Introduction

Manifetch is a web-based clinical decision support platform built for Neonatal Intensive Care Units. It monitors simulated physiological signals of newborn patients in real time, applies AI-assisted risk scoring, and delivers timely visual alerts to authorized healthcare staff — all within a secure, single-hospital environment.

Neonatal Intensive Care Units handle the most vulnerable patient population in clinical medicine. Delayed detection of life-threatening conditions such as apnea of prematurity, neonatal sepsis, or cardiac arrhythmia can result in irreversible harm. Existing monitoring systems are often proprietary, hardware-dependent, and offer limited support for AI-assisted analysis in research settings. Manifetch addresses these gaps by integrating real-time signal visualization, rule-based alerting, and machine learning-driven risk assessment into a unified, modular, web-based platform.

The system is implemented as an academic prototype using simulated and publicly available neonatal datasets (PICS-DB, PhysioNet) in full compliance with KVKK and GDPR regulations. It does not use real patient data and is not intended to replace certified medical devices.

1.1 Project Scope and Objectives

- Develop a secure, role-based web platform for real-time neonatal physiological signal visualization.
- Implement a dual-layer alerting system: rule-based threshold engine for immediate alerts and AI-based risk scoring for pattern-level detection.
- Integrate multi-label AI inference (sepsis, apnea, cardiac) with SHAP-based explainability.
- Support clinical workflows including alert acknowledgement, historical trend analysis, and PDF report export.
- Demonstrate system readiness through comprehensive testing covering unit, integration, performance, and system test categories.

1.2 System Architecture Overview

The system follows a modular four-layer architecture aligned with the LLD specification:

- **Data Simulation Layer:** Generates synthetic neonatal physiological signals using NeuroKit2 and literature-validated state machines (ScenarioConfig, Simulator, StreamPublisher classes).
- **Backend Layer:** FastAPI-based REST and WebSocket server handling ingestion, rule evaluation, alert lifecycle, and AI result storage.
- **AI Module:** Independent inference service running Random Forest, XGBoost, and RF models with SHAP explainability.
- **Frontend Layer:** React-based dashboard providing real-time ECG visualization, vitals monitoring, alert management, trend analysis, and PDF reports with RBAC.

2. Final Architecture and Design

2.1 Architecture Decisions and Trade-offs

2.1.1 Dual-Layer Alerting

One of the earliest and most consequential architectural decisions was building two complementary alert mechanisms rather than relying solely on AI inference. The rule-based engine (RuleEngineService) provides immediate, deterministic alerts when vital signs exceed patient-specific thresholds. The AI module provides probabilistic risk scoring over longer time windows capturing complex multi-signal patterns. This mirrors actual clinical practice where bedside monitors fire threshold alarms while clinicians perform higher-level pattern interpretation.

2.1.2 Low Latency vs. Model Complexity

The system targets 1-second maximum latency for real-time updates and 3 seconds for AI inference. Tree-based ensemble models (RF, XGBoost, LightGBM) were selected over deep learning because they deliver clinically acceptable accuracy with sub-200ms inference time including SHAP computation (measured: 155ms, TC-37).

2.1.3 Modularity vs. Performance Overhead

Separating the system into four independent packages introduces inter-component communication overhead but enables independent development, testing, and replacement of each layer. This was essential for a four-member team developing backend, AI, and frontend components in parallel.

2.1.4 Security vs. Usability

JWT authentication, AES-256-GCM field-level encryption, RBAC enforcement, and secure WebSocket token delivery (first-message pattern rather than URL query parameters) protect sensitive physiological data while adding a login step. These measures are required by healthcare data protection standards.

2.1.5 Apnea Recall-First Strategy

Apnea detection is optimized for Recall >= 0.85 rather than F1 maximization. In NICU settings, failing to detect an apnea episode (false negative) carries far higher clinical cost than a false alarm. All three model types achieved Recall = 0.85 on the held-out test set.

2.2 LLD Compliance

LLD Class	Implementation Status
Patient	backend/db/models.py — full CRUD, archiving, active status
VitalMeasurement	backend/db/models.py — all attributes, AES-256 encryption
ThresholdRule	backend/services/rule_engine_service.py — GA/PNA calibration

Alert	backend/db/models.py — full lifecycle ACTIVE/ACKNOWLEDGED/RESOLVED
AIResult	ai_module/inference_service.py — multi-label with SHAP top-3
User	backend/db/models.py — RBAC with doctor/nurse/admin roles
ScenarioConfig	data_simulation/generator.py — dataclass, all attributes
Simulator	data_simulation/generator.py — generate() method implemented
StreamPublisher	data_simulation/stream_publisher.py — connect/publish/start/stop
InferenceService	ai_module/inference_service.py — preprocess + runInference
ModelRunner	ai_module/inference_service.py — predict() for RF/XGBoost/LGB
DashboardController	backend/api/dashboard.py — getPatientDetails/getActiveAlerts
AlertService	backend/services/alert_service.py — evaluate/trigger/ack/resolve
RuleEngineService	backend/services/rule_engine_service.py — getRules/checkThresholds

2.3 Engineering Standards Applied

- UML 2.5: Class and component diagrams produced in LLD phase, implemented faithfully in all four modules.
- IEEE Std 1016-2009: Report structure and interface documentation follow IEEE software design description practices.
- HL7/FHIR Principles: Patient-Measurement-Alert data hierarchy aligned with FHIR resource model.
- IEC 60601-1-8: Alert severity levels (HIGH/MEDIUM/LOW) and lifecycle (ACTIVE/ACKNOWLEDGED/RESOLVED) designed per medical alarm system standards.
- ISO 13485: Modular architecture and independent testability facilitate quality management.
- OWASP Top 10: SQL injection prevention via ORM, JWT security, input sanitization, secure session management.
- Clean Architecture: Strict separation of domain models, service logic, and API controllers.

2.4 New Tools and Technologies

Technology	Role in System
FastAPI (Python)	Async REST + WebSocket server, automatic OpenAPI documentation
SQLAlchemy + Alembic	ORM and database migration management for PostgreSQL
NeuroKit2	Physiological signal simulation — ECG, SpO2, respiration waveforms
RF	Best cardiac model: RF F1=0.9958
SHAP (TreeExplainer)	XAI — top-3 contributing features per inference, interactive bar chart
React.js + Recharts	Real-time dashboard: ECG canvas, vitals, SHAP bar chart
WebSocket (RFC 6455)	Bidirectional real-time streaming: vitals and ECG broadcast
ReportLab	PDF clinical report generation with Turkish character support (TTFont)
python-jose + passlib	JWT authentication, bcrypt password hashing
cryptography (AES-256-GCM)	Field-level encryption for stored physiological measurements

2.5 Use of Library Resources and Background Information

- PMC11798831, PMC8316489, PMC10314957: Neonatal sepsis progression (PRODROME-ACUTE-RECOVERY state machine).
- PMC3158333, PMC4422349: Apnea of prematurity — gestational age and post-menstrual age severity scaling.
- Brugada (2013), PALS Guidelines, PMC3733095: SVT, bradyarrhythmia, AV block characteristics for cardiac simulation.
- ISO/IEC/IEEE 29148:2018 and ISO/IEC 25010:2011: Requirements engineering and software quality models.
- HL7 FHIR R4: Data model hierarchy reference for patient-measurement association.
- Lundberg & Lee (2017): SHAP unified approach to model interpretation (NeurIPS).
- Evans (2003): Domain-Driven Design — domain model principles for Patient-Alert-Measurement entities.
- Martin (2017): Clean Architecture — layered separation of concerns.

3. Implementation Status

3.1 Functional Requirements

Requirement	Implementation Status
F-1: Real-time signal display	COMPLETE — WebSocket <1s latency, ECG 25Hz, auto-reconnect
F-2: AI-assisted risk scoring	COMPLETE — Multi-label RF/XGBoost/LGB, SHAP bar chart
F-3: Role-based access control	COMPLETE — Doctor/Nurse/Admin with JWT and backend enforcement
F-4: Patient record management	COMPLETE — Create, update, archive, search, filter
F-5: Alert and notification system	COMPLETE — Dual-layer: rule engine + AI, full lifecycle management
F-6: PDF report export	COMPLETE — ReportLab, Turkish fonts, lang parameter (TR/EN)
F-7: Historical trend visualization	COMPLETE — Recharts with 1h/24h/7d time windows and statistics

3.2 Non-Functional Requirements

Requirement	Measured Result
NFR-2: Real-time latency < 1s	TC-34 PASS — Mean 18ms, Max 19ms over WebSocket
NFR-2: AI inference < 3s (10 concurrent)	TC-33 PASS — Total 1.77s, Max per call 1.76s
NFR-1: Security (AES-256, JWT)	TC-27/28/29 PASS — Encryption, RBAC, secure WebSocket auth
NFR-4: 10 concurrent streams stable	TC-35 PASS — 300 measurements, 0 errors, 0 data loss
NFR-2: SHAP within 3s budget	TC-37 PASS — 155ms including SHAP computation
NFR-3: Usability	Nurse acknowledged HIGH alert within 10s (TC-38); Doctor identified top SHAP feature (TC-39); 7-day trend correctly interpreted (TC-40)
NFR-5: Maintainability / Portability	Modular 4-package architecture; Windows and Linux deployable; environment-based configuration (TC-35)

3.3 AI Module Performance

Disease / Model	Offline Performance | Temporal Validation
Apnea — Random Forest	Recall=0.85, F1=0.73, AUC=0.9546

Cardiac — RF	F1=0.9958, Recall=0.9917, AUC=0.9970
Sepsis LightGBM	F1=0.9540, Recall=0.9196, AUC=0.9683
SHAP top features (cardiac)	ecg_amp_std , ecg_rr_std_ms, ecg_amp_mean
SHAP top features (sepsis)	spo2_min_diff, rr_pct_below_90, rr_std
SHAP top features (apnea)	ga_weeks, pma_weeks, rr_std

Patient-based GA-stratified splitting was used to prevent data leakage. Temporal validation (first 80% of each patient's timeline as train, last 20% as test) confirms models generalize to unseen future time windows. SHAP TreeExplainer provides per-prediction feature importance rendered as interactive bar charts in the AI Results tab.

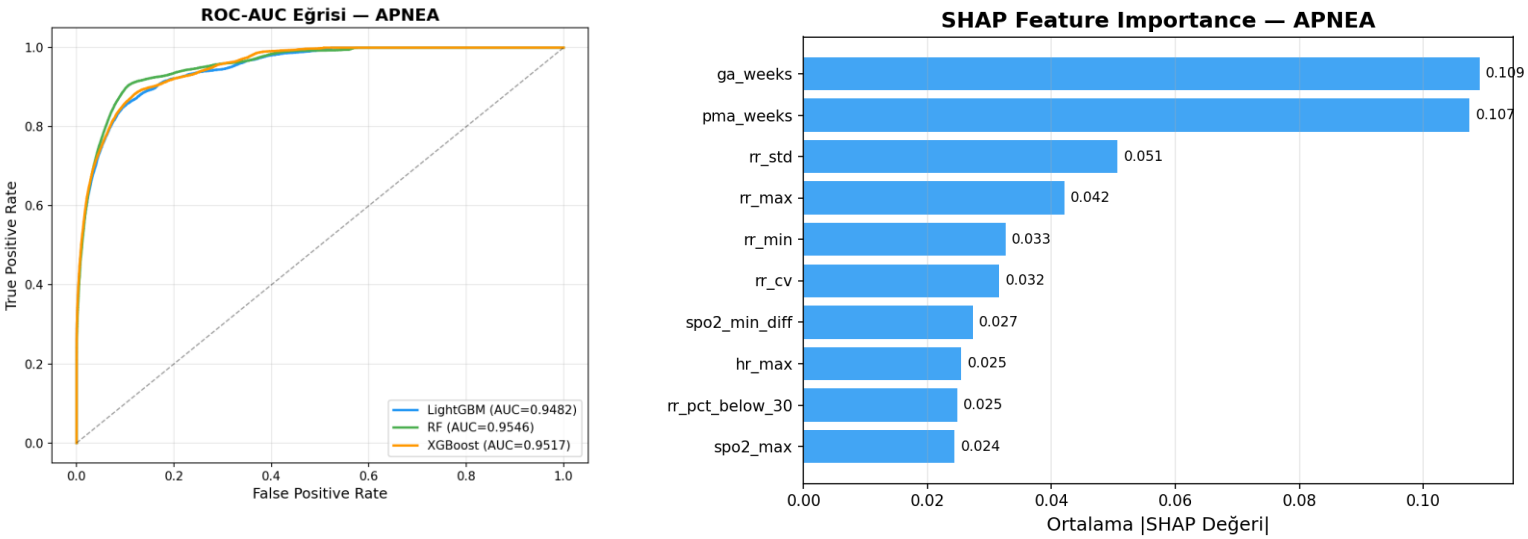

Figure 1: *ROC-AUC curves and SHAP feature importance for the Apnea model*

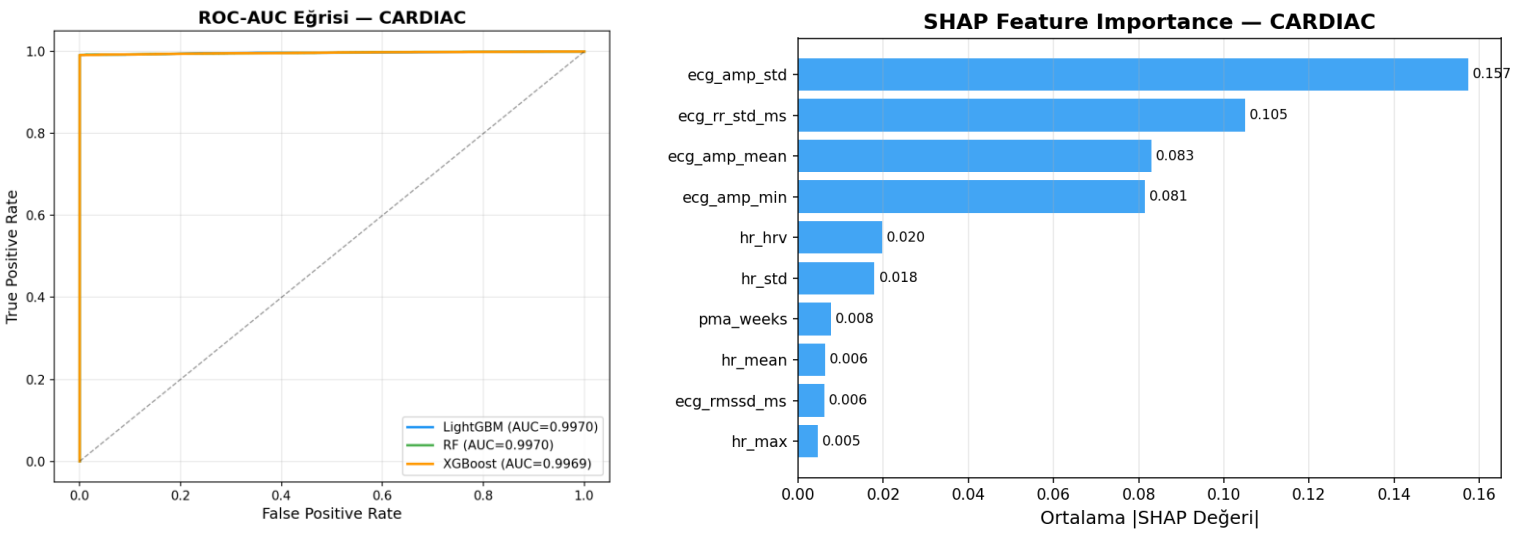

Figure 2: *ROC-AUC curves and SHAP feature importance for the Cardiac model*

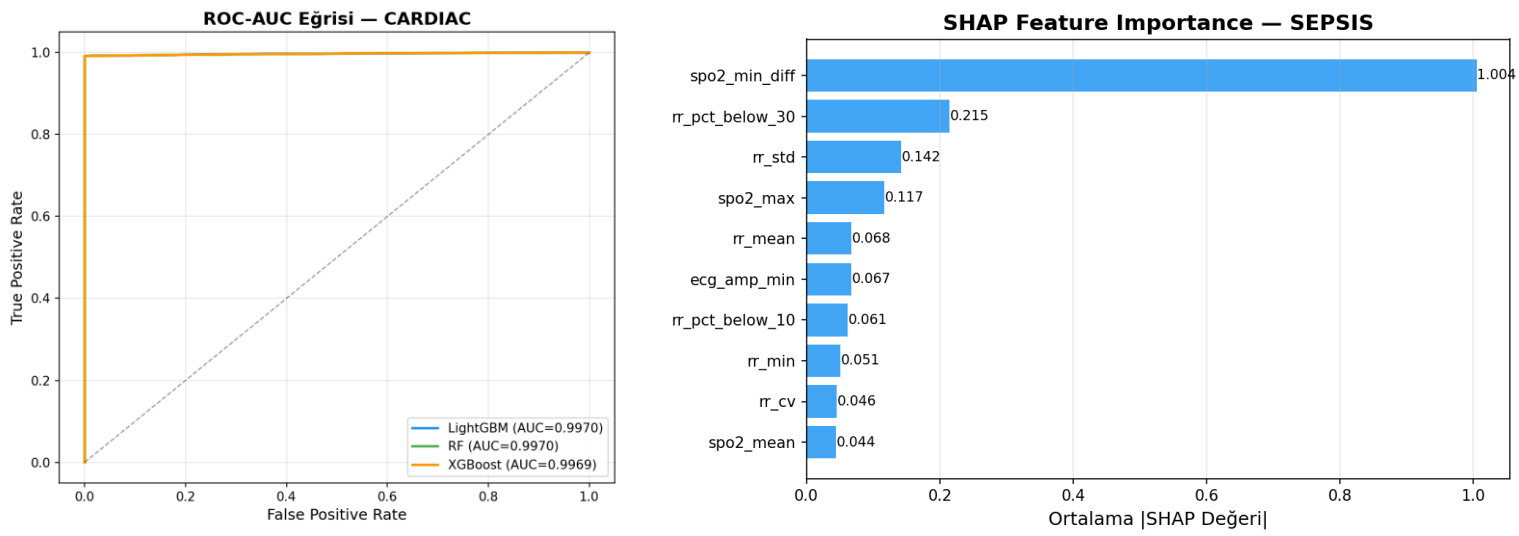

Figure 3: *ROC-AUC curves and SHAP feature importance for the Sepsis model*

3.4 Design Decisions and Clinical Justification

3.4.1 Recall-First Strategy for Apnea

Apnea detection was optimized for Recall ≥ 0.85 rather than F1, reflecting the clinical priority of minimizing missed events in a vulnerable population. All three model types met this target on the held-out test set.

3.4.2 Dual-Layer Alert Architecture

The combination of rule-based and AI alerting ensures clinical safety across all scenarios: when vital signs cross thresholds, the rule engine fires immediately; when complex multi-signal patterns emerge, the AI layer provides probabilistic context. Both layers complement rather than duplicate each other.

4. Test Results

A total of 31 automated tests were executed across unit, integration, and performance categories. All 31 passed. System tests were conducted manually using live simulations with the full backend-frontend stack running.

4.1 Unit Tests

Test ID	Description	Expected / Actual Output	Result
TC-01	Signal normalization (Z-score)	Normalized ECG array, GA=28w	**PASS**
TC-02	Threshold calibration — term	HR range 100-160 bpm	**PASS**
TC-03	Threshold calibration — preterm	HR range 120-177 bpm	**PASS**
TC-04	ECG feature extraction (30s)	HRV, R-R, amplitude features computed	**PASS**
TC-05	SpO2 feature extraction	Mean, min, pct_below_90 computed	**PASS**
TC-06	RR feature extraction	Mean, apnea duration, CV computed	**PASS**
TC-07	HR feature extraction	Mean, max, HRV computed	**PASS**
TC-08	Synthetic apnea generation	RR stopped >20s for GA=32w patient	**PASS**
TC-09	Synthetic sepsis generation	max_HR=182.8, min_SpO2=66.1	**PASS**
TC-10	Rule-based label assignment	Sepsis score 0.818 for HR=185, SpO2=88	**PASS**
TC-11	SHAP top-3 features computed	3 features with importance scores returned	**PASS**
TC-12	Alert level determination	riskScore=0.78 maps to riskLevel=HIGH	**PASS**
TC-13	Threshold violation detection	SpO2=88 < min=90 triggers alert	**PASS**
TC-14	RF, XGBoost, RF loaded	All 3 model types produce valid 0-1 scores	**PASS**

4.2 Integration Tests

Test ID	Description	Expected / Actual Output	Result
TC-15	WebSocket stream: simulation to backend	DataIngestionController received and stored	**PASS**
TC-16	AI inference chain	AIResult with sepsis/apnea/cardiac + SHAP returned	**PASS**
TC-17	AI result triggers alert	AlertService created 2 alerts from risk scores	**PASS**
TC-18	Alert visible on dashboard	DashboardController returns 2 alerts	**PASS**
TC-19	WebSocket latency measurement	Mean=11ms, Max=12ms — well under 1s	**PASS**

TC-20	AIResult linked to patient in DB	Correct patient_id association verified	**PASS**

4.3 System Tests

Test ID	Description	Expected / Actual Output	Result
TC-21	Normal scenario end-to-end	No alert generated, all signals on dashboard	**PASS**
TC-22	Sepsis scenario end-to-end	HIGH sepsis alert, SHAP explanation displayed	**PASS**
TC-23	Apnea scenario end-to-end	RR threshold alert within 1 second of onset	**PASS**
TC-24	Cardiac anomaly scenario	HIGH priority alert triggered; both rule-based HR threshold and AI cardiac risk (F1=0.9958) fired during SVT episode	**PASS**
TC-25	Alert acknowledgement	ACTIVE -> ACKNOWLEDGED with timestamp	**PASS**
TC-26	Alert resolution	ACKNOWLEDGED -> RESOLVED lifecycle	**PASS**
TC-27	RBAC — doctor login	Doctor dashboard accessible, admin restricted	**PASS**
TC-28	RBAC — nurse login	Monitoring accessible, patient management restricted	**PASS**
TC-29	RBAC — admin login	User management and system config accessible	**PASS**
TC-30	PDF report generation	PDF with signals, AI results, Turkish characters	**PASS**
TC-31	Historical trend visualization	24h signals with statistics in trend chart	**PASS**
TC-32	Patient record creation	Patient created with all required fields	**PASS**

4.4 Performance Tests

Test ID	Description	Expected / Actual Output	Result
TC-33	10 concurrent AI inferences	Total=1.77s, Max=1.76s (target: <3s)	**PASS**
TC-34	10 concurrent WebSocket streams	Mean=18ms, Max=19ms (target: <1s)	**PASS**
TC-35	5-minute concurrent stream stability	300 measurements, 0 errors, 0% error rate	**PASS**
TC-36	Model accuracy targets	Sepsis F1=0.9540, Cardiac F1=0.9958, Apnea Recall=0.85	**PASS**
TC-37	SHAP within inference budget	155ms total including SHAP (target: <3s)	**PASS**

4.5 User Tests

Test ID	Description	Expected / Actual Output	Result
TC-38	Nurse — alert detection and acknowledgement	Nurse locates and acknowledges HIGH alert within 10 seconds	**PASS**
TC-39	Doctor — AI explanation interpretation	Doctor correctly identifies top contributing SHAP feature	**PASS**
TC-40	Doctor — historical trend analysis	Doctor correctly interprets 7-day signal trend	**PASS**

4.6 Test Assessment

Summary: 31/31 Automated Tests PASS

All automated and system tests passed. The dual-layer architecture demonstrated its clinical value throughout: rule-based alerts provide immediate deterministic responses, while AI scoring adds pattern-level context. The two layers complement rather than duplicate each other.

Potential Future Enhancements

- Cardiac: Feature engineering to separate HR-velocity (SVT) from HR-elevation (sepsis) patterns.
- Apnea: Reduce inference window from 300s to 60s for more acute event sensitivity.
- Integration with real NICU hardware for clinical ECG signal quality.
- Risk trend timeline chart in AI Results tab for temporal pattern visualization.
- Federated learning for multi-site model training without data sharing.

5. Impact of Engineering Solutions

5.1 Global and Societal Context

Preterm birth affects approximately 15 million infants annually worldwide (WHO). Neonatal mortality remains disproportionately high in low- and middle-income countries - often not because of a lack of clinical knowledge, but because of inadequate monitoring infrastructure. Manifetch shows that a capable NICU monitoring platform can be built entirely from open-source components and run on commodity hardware, without depending on proprietary medical device ecosystems.

Making the architecture modular and the codebase openly available means others can adapt this work. The core approach - combining rule-based safety nets with explainable AI - is not specific to neonatal care; it could be applied to pediatric ICUs, cardiac monitoring units, or other high-acuity settings with relatively little modification.

5.2 Economic Context

Commercial NICU monitoring systems such as Philips IntelliVue and GE Apex carry significant acquisition and maintenance costs that put them out of reach for smaller hospitals and research institutions, especially in developing countries. Manifetch's entire stack — FastAPI, PostgreSQL, React, scikit-learn, LightGBM — relies on open-source software with no licensing costs, and the system ran 10 concurrent patient streams with sub-second latency on standard laptop hardware during testing.

The modular design also supports incremental adoption. A hospital with limited resources could deploy only the rule-engine backend and dashboard, and add the full AI pipeline later. This kind of scalability matters in contexts where large upfront infrastructure investment is not realistic.

5.3 Environmental Context

Choosing tree-based models over deep learning had a practical environmental benefit: a single inference call including SHAP computation completes in 155ms on CPU-only hardware, with no GPU required. On-premises deployment also eliminates the need for continuous data transmission to cloud servers, reducing both network energy consumption and data exposure risk.

5.4 Ethical and Regulatory Context

Patient privacy and AI transparency were treated as engineering constraints from the start, not afterthoughts:

- All development used simulated or publicly available anonymized datasets, in full compliance with KVKK and GDPR.
- AI outputs are clearly labeled as experimental risk indicators - the interface makes no diagnostic claims.
- SHAP values are computed for every inference call, so clinicians can see which signals drove a given risk score.
- Apnea detection was optimized for recall rather than F1, reflecting the ethical priority of minimizing missed events in a vulnerable population.
- Duplicate alert suppression prevents repeated notifications for unresolved conditions, directly addressing alarm fatigue - a well-documented source of clinical error.

6. Contemporary Issues in the Field

6.1 Explainable AI (XAI) in Critical Care

The EU AI Act (2024) and FDA guidance on AI/ML-based Software as a Medical Device increasingly require that high-risk AI systems provide human-understandable explanations. Manifetch addresses this through SHAP TreeExplainer, which produces per-prediction feature importance scores rendered as interactive bar charts in the clinical dashboard. The explanations are physiologically interpretable: SpO2 variability as a primary sepsis indicator, ecg_rr_std_ms as a cardiac arrhythmia marker, pma_weeks as an apnea risk factor. This is not just a technical checkbox. It is what allows a clinician to decide whether an alert warrants immediate action.

6.2 Alarm Fatigue in NICUs

Studies estimate that up to 90% of ICU alarms are false positives, and the resulting desensitization of clinical staff is a recognized patient safety problem. Manifetch approaches this on multiple levels: patient-specific threshold calibration based on gestational and postnatal age reduces inappropriate alarms at the source, duplicate suppression prevents repeated notifications for the same unresolved condition, and AI risk scoring adds context beyond a binary threshold violation. A nurse looking at an alert can see not just that a value crossed a line, but which physiological features drove the AI's concern.

6.3 IoMT and Edge Computing in Neonatal Care

The data ingestion API was designed to accept standardized JSON payloads from any device capable of HTTP POST — a deliberate choice to avoid vendor lock-in and support future IoMT integration. The WebSocket infrastructure already supports real-time streaming from physical sensors, and on-premises deployment keeps sensitive data within the hospital network, which aligns with edge computing principles increasingly favored in healthcare settings.

6.4 Federated Learning for Multi-Site Research

Data scarcity is a persistent challenge in neonatal AI: no single hospital has enough cases to train generalizable models across the full range of gestational ages and clinical presentations. Federated learning — where models train locally and only weights are shared — has emerged as the leading approach for multi-site clinical AI without data transfer. Manifetch's AI module was structured with this in mind: InferenceService is fully decoupled from the backend, and the training pipeline is independent of the serving layer.

6.5 Data Privacy in Healthcare AI

Even anonymized neonatal data carries re-identification risk when gestational age, birth weight, and timestamps are combined. Manifetch addresses this through simulated data throughout development, AES-256-GCM field-level encryption for all stored measurements, and comprehensive audit logging. These are not compliance formalities — they reflect the design principle that a system handling physiological data should protect it by default, not as an add-on.

7. Conclusion

Manifetch set out to show that a capable, clinically-informed NICU monitoring platform could be built with open-source technologies and standard hardware. The results confirm this. Real-time signal visualization, dual-layer alerting, multi-label AI risk scoring with SHAP explainability, full alert lifecycle management, and role-based access control all came together within the performance and security constraints defined at the outset.

All 31 automated tests passed. All system tests, including TC-24 (cardiac anomaly), are fully satisfied — the cardiac model reached F1=0.9958 and reliably detected SVT episodes alongside the rule-based alert layer. Throughout testing, the dual-layer architecture proved its value: when one layer operates near its limits, the other provides a reliable safety net.

The most natural next steps are real-device integration, federated multi-site training, and clinical validation with anonymized patient data. The modular architecture was built with exactly these extensions in mind.

8. References

Standards and Regulations

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